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# A sample wetting strategy to overcome differences in physical morphology between lab-prepared training samples and pharmaceutical process samples for reliable quantitative Raman spectroscopic analysis

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As a training set for multivariate quantitative analysis, lab-blended powder samples are frequently used due to the facile concentration variation of the individual components. Real pharmaceutical process samples, however, are often granular in form. The different morphologies between training and process samples result in dissimilar Raman spectral features, thereby seriously deteriorating the accuracy of the analysis. To overcome this hurdle, an effective and simple strategy to make physical presentations of both training and process samples similar, called water wetting, was demonstrated in this study. Wetting potentially dismantles process granule samples into smaller aggregates, whereas lab powder samples could be somewhat aggregated by water-bridging. Thus, water wetting would produce similar morphologies of both samples. To evaluate the strategy, samples containing esomeprazole magnesium dihydrate (EMD) as an active pharmaceutical ingredient (API) and five additional components were employed. Initially, training samples were mixed with the individual components at the intended concentration ranges. A partial least squares (PLS) model developed using the Raman spectra of the training samples was inaccurate in determining EMD concentrations in the real process samples. When both training and process samples were wetted prior to analysis, accurate concentration determination was realized due to no significant difference in Raman spectral features between samples. The similar morphologies and crystallinities of the two types of wetted samples were confirmed by analyzing the corresponding SEM images and XRD patterns. The proposed strategy is versatile and expandable for further vibrational spectroscopic analysis of various other samples with different morphologies.

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## 1. Introduction

Raman spectroscopy has been frequently employed as a fast and non-destructive analytical tool for the determination of active pharmaceutical ingredient (API) concentration in diverse pharmaceutical samples.[1–7] Multivariate chemometric methods[3,7–9] are mostly adopted not only for Raman spectroscopy, but also for near-infrared (NIR) spectroscopic analysis to determine the concentration of a target component in sample mixtures, and partial least squares (PLS) is the most routine choice.[4,5] To build a robust multivariate model for compositional analysis, the most demanding prerequisite is the preparation of a training sample set, incorporating wide concentration variations of individual components without concentration correlations among the components, namely the absence of concentration collinearity, to ensure the component-specificity of a model.[10,11] The most facile route to realize this goal is by the preparation of training samples by blending individual components at designed concentration ranges, while taking care to avoid concentration correlations.[9,12]

Another important issue to consider is whether the physical characteristics of training samples are sufficiently representative of those of target validation samples, such as real process samples. Generally, many samples in real pharmaceutical process steps are in a granular form,[13–18] whereas training samples are most likely prepared with non-granular powders

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of each component. As is well known, the Raman spectral features of samples vary sensitively to morphology and crystallinity.[5,19] Therefore, the dissimilar morphologies between training and process samples result in different Raman spectral features even when the compositions of the samples are the same, so ultimately leading to substantial deterioration of prediction accuracy. One straightforward way to overcome this problem is the direct use of granular process samples as training samples; however, the concentrations of ingredients in process samples vary minimally within the limits of quality control. Process samples with such narrow component concentration variations would not be adequate as training samples due to lack of linearity[20] and existence of concentration collinearity. Therefore, a subsequent multivariate analysis would not be robust. Addition of other components into a process sample to increase concentration variation would be possible; however, controlling the morphologies of the added samples similar to that of a process sample is practically unachievable.

Attaining similar physical presentations of both training and process samples is the only method to realize accurate compositional analysis by Raman spectroscopy. A simple strategy of sample wetting by water is suggested in this study.[21] The water wetting potentially dismantles granules in a process sample into smaller piece-like aggregates,[22] while a lab powder sample would be somewhat aggregated by water-bridging.[23] By this method, the morphologies of the two samples would become more similar, and the morphology-induced Raman spectral differences would be minimized. The practice proposed is simple and facile, involving only addition of water into a sample.

To evaluate the utility of the proposed scheme, pharmaceutical samples (referred to as EMD samples in this study) composed of esomeprazole magnesium dihydrate (EMD), L-arginine, povidone, cellulose, MgO, and sodium stearyl fumarate (SSF) were employed in this study. EMD was used as an API. Initially, training samples were prepared by mixing the individual components at the intended concentration ranges and confirming no concentration correlations among the components. Using the Raman spectra of training samples, a PLS model was developed. This model was used to predict EMD concentrations in the process samples composed of granular forms, and the resulting accuracy was evaluated. In parallel, differences in Raman spectral features between the training and process samples were examined.

Next, both training and process samples were wetted with water and subjected to Raman analysis. The varied spectral features resulting from the wetting process were examined and compared with those before wetting. Another PLS model was built using the collected training sample spectra, and the resulting accuracy was evaluated in parallel with that obtained using the dry (non-wetted) samples. To investigate the changes of morphology and crystallinity after wetting and to explain the subsequent varied Raman spectral features, X-ray diffraction (XRD) and scanning electron microscopy (SEM) images of both dry and wetted samples were acquired, and their characteristics were analyzed.[24,25] Finally, expandability of the wetting strategy for the analysis of other samples was discussed.

## 2. Experimental section

### 2.1 Sample preparation and sample wetting

All of the samples (30 training and 6 process samples) used in this study were supplied by Chong Kun Dang Pharmaceutical Co., Cheonan, Korea. As mentioned, the samples were composed of 6 ingredients: EMD (9.6 wt%), L-arginine (30.0 wt%), povidone (0.4 wt%), cellulose (43.0 wt%), MgO (15.0 wt%), and SSF (2.0 wt%). Povidone and SSF were a bonding agent and a glidant, respectively. L-Arginine, cellulose, and MgO were excipients. The EMD concentration for each sample was expressed as the percentage relative to the amount of EMD in a final tablet (21.7 mg). EMD concentrations in the training samples varied from 81.5% to 119.3%, and no concentration correlations among components were confirmed. The concentration ranges of L-arginine and cellulose were 80.9–119.4% and 81.6%–113.9%, respectively. Six process samples were also independently acquired during separate batches of the actual tablet manufacturing process.

For wetting of a sample, 1 mL of deionized water was added to 1.0 g of sample in a rectangular cuvette with a thickness of 3.0 mm. The water-added sample was mixed thoroughly using a thin wire over 5 minutes. The final sample had the consistency of a thin paste. Next, the sample-containing cuvette was tapped several times to eliminate potential bubbles and then placed under laser radiation for spectral acquisition. This procedure was accomplished over 5 minutes for all the samples to maintain consistency of wetting.

### 2.2 Acquisition of Raman spectra, XRD diffractograms, and SEM images of samples

Raman spectra of samples were collected using a transmission Raman spectrometer (Cobalt Light Systems) operating with a 830 nm laser (power: 650 mW). The size of the laser illumination spot was 4 mm, and the exposure time was 10 seconds. Triplicate spectra were recorded for each sample at a resolution of 8 $cm^{-1}$. All pretreatments of Raman spectra and multivariate analyses, including baseline correction, intensity normalization, principal component analysis (PCA), and PLS, were conducted using MATLAB version 2017a (The Math Works Inc., MA, USA).

XRD diffractograms of samples were obtained using an X-ray diffractometer (Rigaku Model SmartLab, Rigaku Corporation, USA) under Cu Kα X-ray radiation operated at 45 kV and 200 mA. The incidence angle was varied from 3 to 45° with an increment of 0.02°, and the scan rate was 2° $min^{-1}$. To obtain SEM images (Hitachi S-4800, Japan), Pt coating was applied on the surface of a sample using a Pt sputter (Hitachi E-1045, Japan) at a rate of 8–10 nm $min^{-1}$. A wetted sample was homogeneously spread on a glass plate for XRD measurement under ambient conditions. During the measurement, wetted water was slightly evaporated; nonetheless, no significant changes in XRD features were observed.

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# 3. Results and discussion

## 3.1 Raman spectral features of individual components and mixture samples

Fig. 1 shows the transmission Raman spectra (1400–900 $cm^{-1}$ range) of all individual components of the EMD samples. The peaks of each component are considerably distinct from each other, and the peaks of API (EMD) are relatively intense, with the strongest band at 1249 $cm^{-1}$, corresponding to the C–C vibration. The 1362 and 1001 $cm^{-1}$ bands arise from the $CH_3$ bending and the C–C aromatic ring chain vibration, respectively.[26–29] The peak intensities of SSF are similar to those of EMD, and the peaks of L-arginine are most intense, while the peaks of povidone, MgO, and cellulose are weak and somewhat broad. The peaks of the components overlap in the spectra of mixture samples; so, multivariate calibration methods are necessary to recognize the concentration variation of EMD, and the most intense EMD band at 1249 $cm^{-1}$ is greatly informative for this purpose.

Fig. 2(a) shows the Raman spectrum of a lab-mixture training sample. The weight percents of EMD, cellulose, L-arginine, MgO, SSF, and povidone in the sample were 9.6, 43.0, 30.0, 15.0, 2.0, and 0.4%, respectively. Cellulose and L-arginine are the two most dominant components. As discussed, the spectral features result from the overlapping peaks of the individual components at their corresponding concentrations. So, many diverse peaks are present in the spectrum, with the most intense bands at 1249 and 983 $cm^{-1}$. Although the concentration of cellulose is the highest, the corresponding peak in the spectrum is difficult to find since it is very weak and broad (refer to Fig. 1). The most intense band at 983 $cm^{-1}$ is mainly from L-arginine due to the second highest concentration of L-arginine. The 1249 $cm^{-1}$ band is contributed largely by EMD, although its concentration is not high at only 9.6%. Since the two most dominant components (cellulose and L-arginine) do not have strong bands around 1249 $cm^{-1}$, this EMD band is prominent.

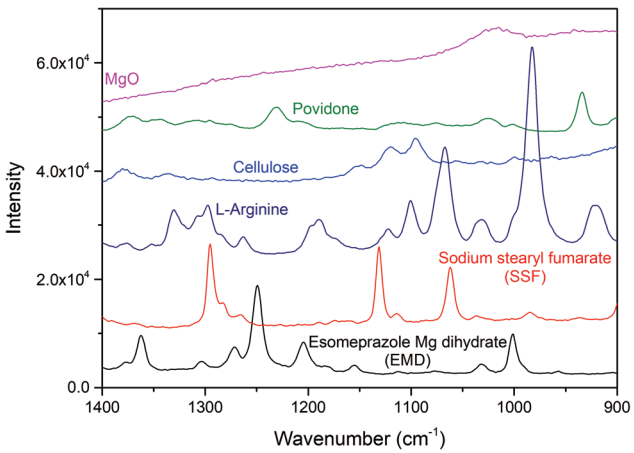


Fig. 1 Transmission Raman spectra of pure components of the EMD sample in the 1400–900 $cm^{-1}$ range. Individual spectra are arbitrarily offset for clear comparison.

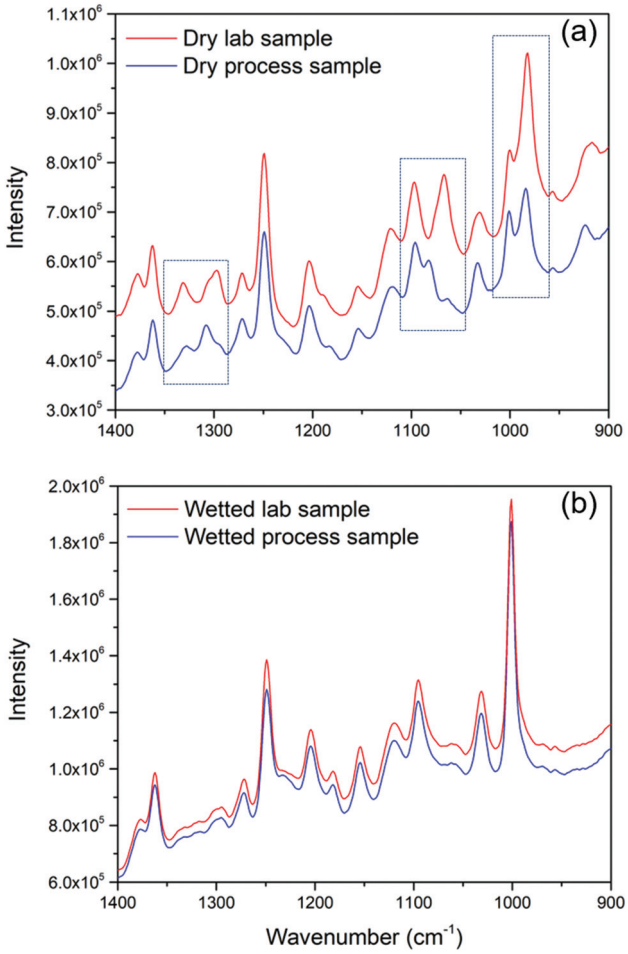


Fig. 2 Raman spectra of dry (a) and wetted (b) lab and process samples (not baseline-corrected).

The Raman spectrum of a process sample possessing the same component concentrations as the lab sample is also shown in Fig. 2(a). At a glance, the spectral features of the lab and process samples are clearly different, especially in the 1348–1282, 1110–1046, and 1014–962 $cm^{-1}$ ranges (marked by the boxes). The band positions and shapes in these ranges are clearly dissimilar from each other. Based on the observation, it is apparent that a multivariate model developed using the lab-blended samples will not accurately predict EMD concentrations in the process samples.

For easy examination of all spectral features of the lab and process samples together, PCA was performed using the corresponding spectra in the 1400–900 $cm^{-1}$ range, and the resulting scores were evaluated. Fig. 3(a) shows the first and second scores corresponding to all the lab (red circles) and process (blue circles) samples in a two-dimensional domain. The first and second eigenvectors described 95.9% and 2.9% of the total variation in the sample dataset. As shown, the scores of the process samples were located substantially distant from the scores of lab samples. This result clearly indicates the different spectral features between the lab and process samples. The origin of the difference must be addressed.

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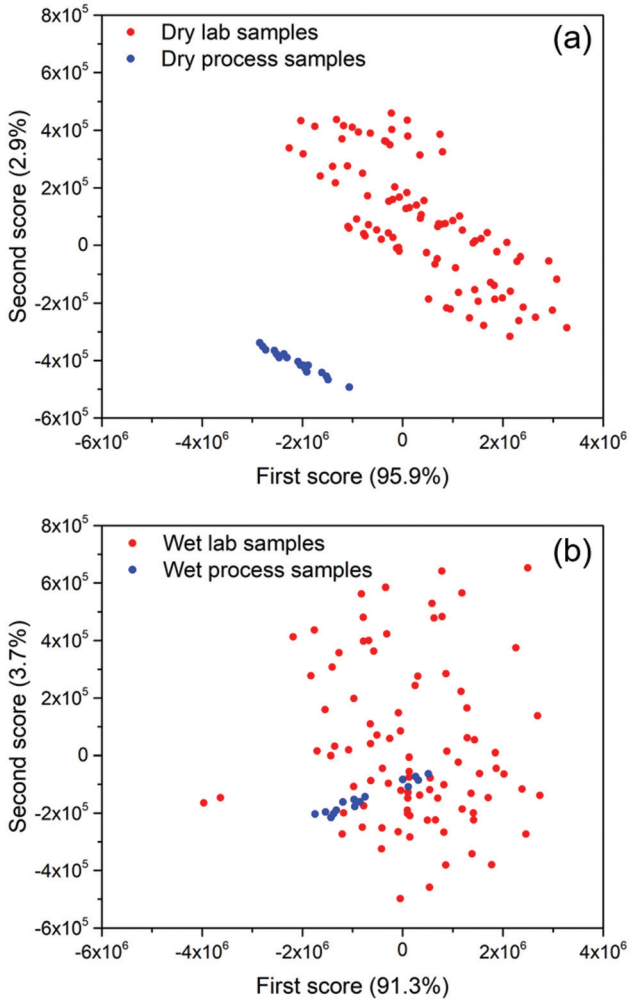


**Fig. 3** Two-dimensional scatter plots of the 1st and 2nd scores obtained from the spectra of all dry (a) and wetted (b) samples. The numbers in parentheses indicate the percentages of spectral variation described by the respective eigenvectors relative to the total spectral variation in the dataset.

### 3.2 Examination of the crystalline features of individual components and mixture samples by XRD

Since the crystallinity of a sample influences its Raman spectral features, crystalline characteristics of the lab and process samples were initially examined using XRD analysis. Before moving to the analysis of mixture samples, XRD diffractograms of individual components were acquired, and their diffraction patterns were examined, as shown in Fig. 4. In the cases of EMD, L-arginine, and SSF, many narrow and sharp peaks are apparent, thereby indicating crystalline structures with varied sizes of crystal units. MgO is also crystalline with fairly uniform crystallinity, and cellulose is semi-crystalline, also containing amorphous structures, based on the observation of relevant XRD peaks. The XRD pattern of povidone clearly indicates that it is amorphous.

Fig. 5 shows XRD diffractograms of the same dry lab and process samples. The overall XRD features of both samples are generally similar, while the detailed peak shapes are different, especially in the 17.5–20.0 and 26.0–29.5° ranges. For quantitative evaluation, the relative crystallinity of each sample was calculated based on the Ruland–Vonk method,[30,31] which obtains the percentage area of the crystalline diffraction relative to the total area of the diffractogram (refer to the equation below). The total area corresponds to the sum of crystalline and amorphous contributions. Before calculation, each diffractogram was baseline-corrected and smoothed to decrease noise.

$$x_{\mathrm{CR}}(\%) = \left( \int_{s_o}^{s_p} I_{\mathrm{CR}}(s) \cdot s^2 \mathrm{d}s \Big/ \int_{s_o}^{s_p} I(s) \cdot s^2 \mathrm{d}s \right) \times 100$$

In the equation, $I_{\mathrm{CR}}(s)$ is the intensity of the crystalline peak, and $I(s)$ is the summed intensity of the crystalline and amorphous peaks. The relative crystallinity ($x_{\mathrm{CR}}(\%)$) of the dry lab and process samples was 65.5 and 60.6%, respectively. Since the process sample was produced by wet granulation using a binder (aqueous povidone solution) including spraying and moistening (liquid bridging of particles) steps, the overall crystallinity potentially decreased, and the crystalline structures of the components likely varied, resulting in the observed different XRD patterns.

Meanwhile, the decreased crystallinity of the process sample was not significant. Therefore, the different Raman spectral features shown in Fig. 2(a) were attributed in part to their dissimilar crystallinities; however, this is not the sole source of the differences. The granular morphology of the process sample would result in different Raman spectral features since the morphology of a sample is one of the key factors influencing Raman peak shapes. Overall, it would be reasonable to conclude that the observed dissimilar Raman spectral features originated from the differences in both overall crystallinity and morphology between the lab and process samples.

### 3.3 Wetting of lab and process samples and examination of their Raman spectral features

Since the morphologies of lab and process samples are expected to be different, sample wetting was proposed as a strategy for producing similar sample morphologies, resulting in similar Raman spectral features. The wetting of a sample was performed as described in the Experimental section. Fig. 2(b) shows the Raman spectra of the wetted lab and process samples. Two noticeable observations are apparent in the spectra. First, the number of peaks in the spectral range decreased, and overall spectral features became simpler. This indicates disassembly of granular forms and reduced crystalline structure of the samples by wetting. Second, overall band shapes of the two samples became nearly identical, indicating very similar physical conditions (presentations) of the samples.

For examination of the spectral features of all the lab and process samples after wetting, PCA was again performed using the corresponding spectra. Fig. 3(b) shows the two-dimensional scatter plot of the first and second scores obtained from

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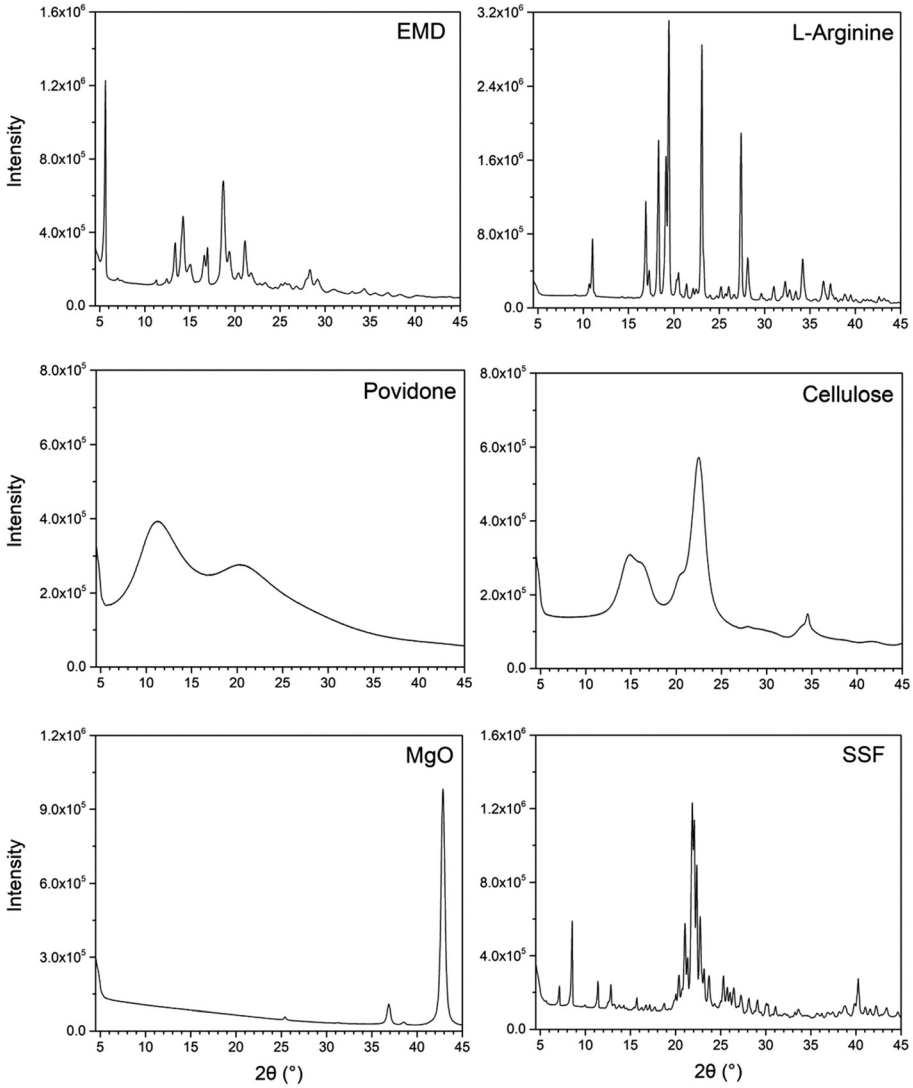


Fig. 4 XRD diffractograms of pure components in the EMD sample.

the spectra of all wetted samples. Now, the blue scores (process samples) were within the group of red scores (lab samples), indicating similar spectral features resulting from analogous morphologies and crystallinities of the samples. In addition, the first and second eigenvectors described 91.3% and 3.7% of the total variation in the wetted sample dataset, respectively. The decreased description by the first eigenvector indicates that the spectral features of the wetted samples were less component-characteristic. The observations of the spectra and the subsequent score scatter plot support the feasibility of accurate prediction of the EMD concentration in process samples by using a multivariate model developed using wetted lab samples.

### 3.4 Examination of physical characteristics of wetted samples by SEM and XRD

It was necessary to provide acceptable rationales for the similar spectral features of lab and process samples. For this purpose, SEM images of the samples were acquired to examine their morphologies (refer to Fig. 6). The SEM image of the dry lab sample (Fig. 6(a)) reveals that the particle shape is quite homogeneous and sharply rod-like with high crystallinity, whereas the dry process sample (Fig. 6(b)) comprised granules of much larger sizes (~50–200 μm), with the absence of rod-like particles. The morphologies of the dry lab and dry process samples were completely different, indicating their dissimilar Raman spectral features. When the lab sample was wetted, the morphology changed dramatically due to the aggregation of individual particles by physical adhesion due to water-bridging, as shown in Fig. 6(c). Small aggregates with varying sizes were clearly observed along with occasional large aggregates (~25–40 μm). The wetting of the process sample changed the sample morphology (Fig. 6(d)) into much smaller aggregates from granules. The dismantling of the granular structure by wetting was the cause, as discussed previously. As observed in the SEM images, wetting produced more similar morphologies of the two samples.

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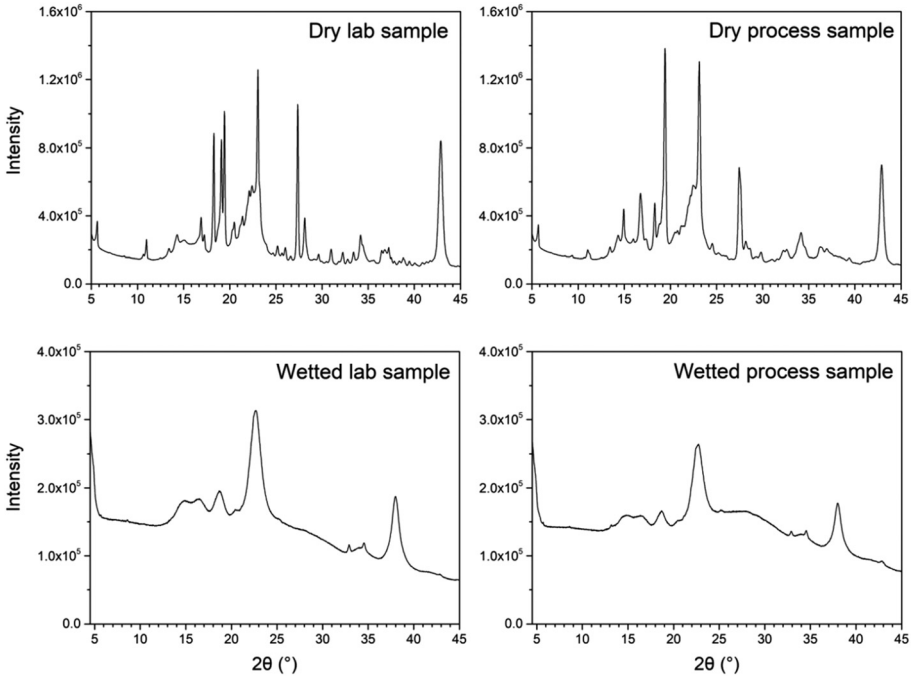


Fig. 5 XRD diffractograms of the dry lab, dry process, wetted lab, and wetted process samples.

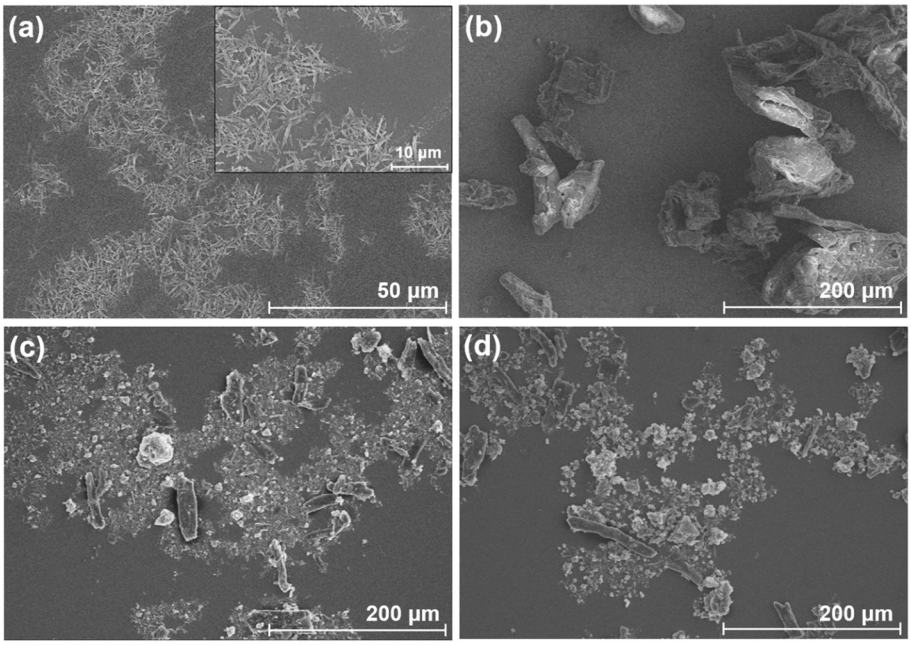


Fig. 6 SEM images of the dry lab (a), dry process (b), wetted lab (c), and wetted process (d) samples.

To assess the potential changes in crystallinity after wetting, XRD diffractograms of the two wetted samples were acquired (refer to Fig. 5). Wetting resulted in dramatically simpler and broader XRD patterns of both samples, indicating the disappearance of a large portion of crystallinity through contact with water. Simultaneously, the XRD patterns became similar after wetting. The calculated relative crystallinities of the wetted lab and wetted process samples were 17.0 and 16.8%, respectively, very similar to each other. Wetting dramatically decreased the crystallinities of the two samples in comparison with the dry samples. Overall, wetting was effective in producing similar morphologies and crystallinities of lab and process samples, so that similar Raman spectral features were obtained.

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### 3.5 Determination of API concentrations in the process samples

To determine the EMD concentrations in the process samples, PLS was employed. Raman spectra of the original dry and wetted lab samples in the 1400–900 $cm^{-1}$ region were separately used for PLS. Each PLS model was constructed by leave-one-out cross-validation. Table 1 shows the results of PLS calibrations using both dry and wetted samples, and Fig. 7 shows the corresponding concentration correlation plots. The red and blue circles correspond to the training (lab) and prediction (process) samples, respectively. As presented, the standard errors of cross-validation (SECVs) in the two cases were similar, although their Raman spectral features were different. It seemed that the more detailed Raman bands of the dry sample would be helpful for tracking the concentration variation of EMD; however, EMD-relevant information is comparable in the two spectra regardless of morphology and crystallinity. This explains the similar SECVs in PLS modeling.

Using each PLS model, the EMD concentrations of 6 process samples were predicted, and the resulting accuracy was then evaluated. The model developed using the dry lab samples, as expected, could not predict the EMD concentrations accurately. With a standard error of prediction (SEP) of 29.47%, the predicted concentrations were far from the calibration line. In contrast, when both lab and process samples were wetted, the EMD concentrations were measured with good accuracy (SEP: 1.68%). Adjustment of the physical morphologies and crystallinities of the samples to similar levels was the origin of the accurate concentration determination using Raman spectroscopy.

**Table 1** PLS accuracies resulted in the determination of EMD concentrations of dry and wetted samples for Raman spectral acquisition

| Samples used for training | Number of factors | SECV (%) | Samples used for prediction | SEP (%) |
|---|---|---|---|---|
| Dry lab samples | 6 | 1.99 | Dry process samples | 29.47 |
| Wetted lab samples | 6 | 2.05 | Wetted process samples | 1.68 |

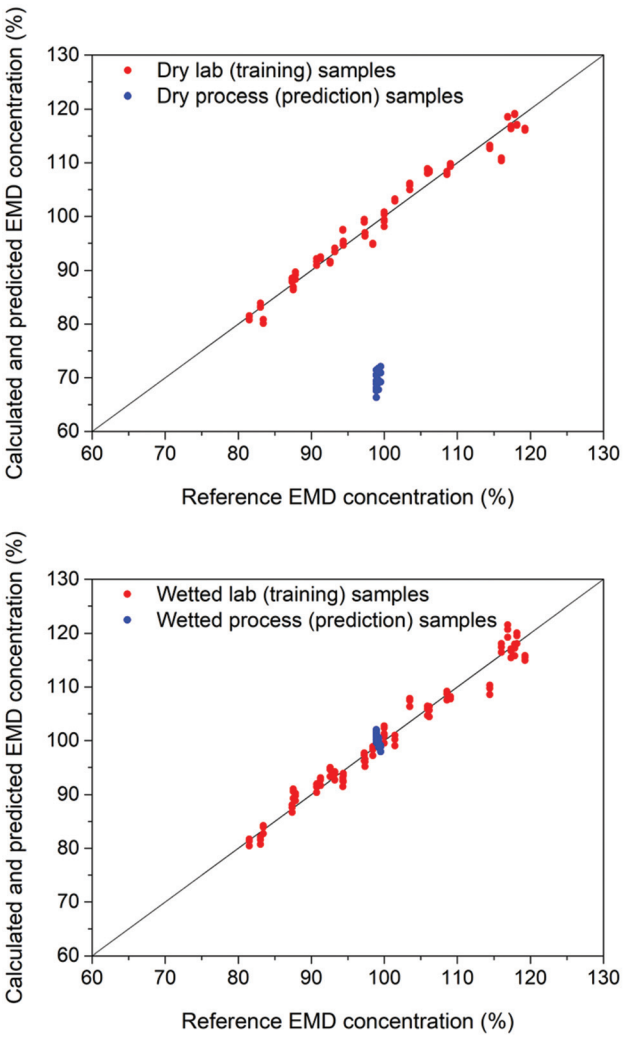


**Fig. 7** Concentration correlation plots for the determination of EMD concentration by PLS, when dry (top) and wetted (bottom) samples were used for analysis. The red and blue circles correspond to the training (lab) and prediction (process) samples, respectively.

## 4. Conclusion

Water wetting was a simple and effective strategy to produce similar morphologies and crystallinities of lab training and process samples. It enabled the development of a PLS model using lab-blended samples to predict API concentrations in process samples without the use of further spectral treatment algorithms. The overall duration of wetting and spectral acquisition was less than 10 minutes; therefore, it is practically applicable in routine analysis. The proposed strategy is beneficial when the accurate multivariate spectroscopic analysis is unrealizable due to differences in morphologies between training and process samples. This situation occurs frequently; therefore, we think that the wetting method will be adopted for the analysis of diverse pharmaceutical and agricultural powder samples. Future research will be conducted to expand the method to determine API concentrations in different pharmaceutical samples with various morphologies and to explore optimal wetting agents in accordance with the physical characteristics of a selected sample.

## Conflicts of interest

There are no conflicts to declare.

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